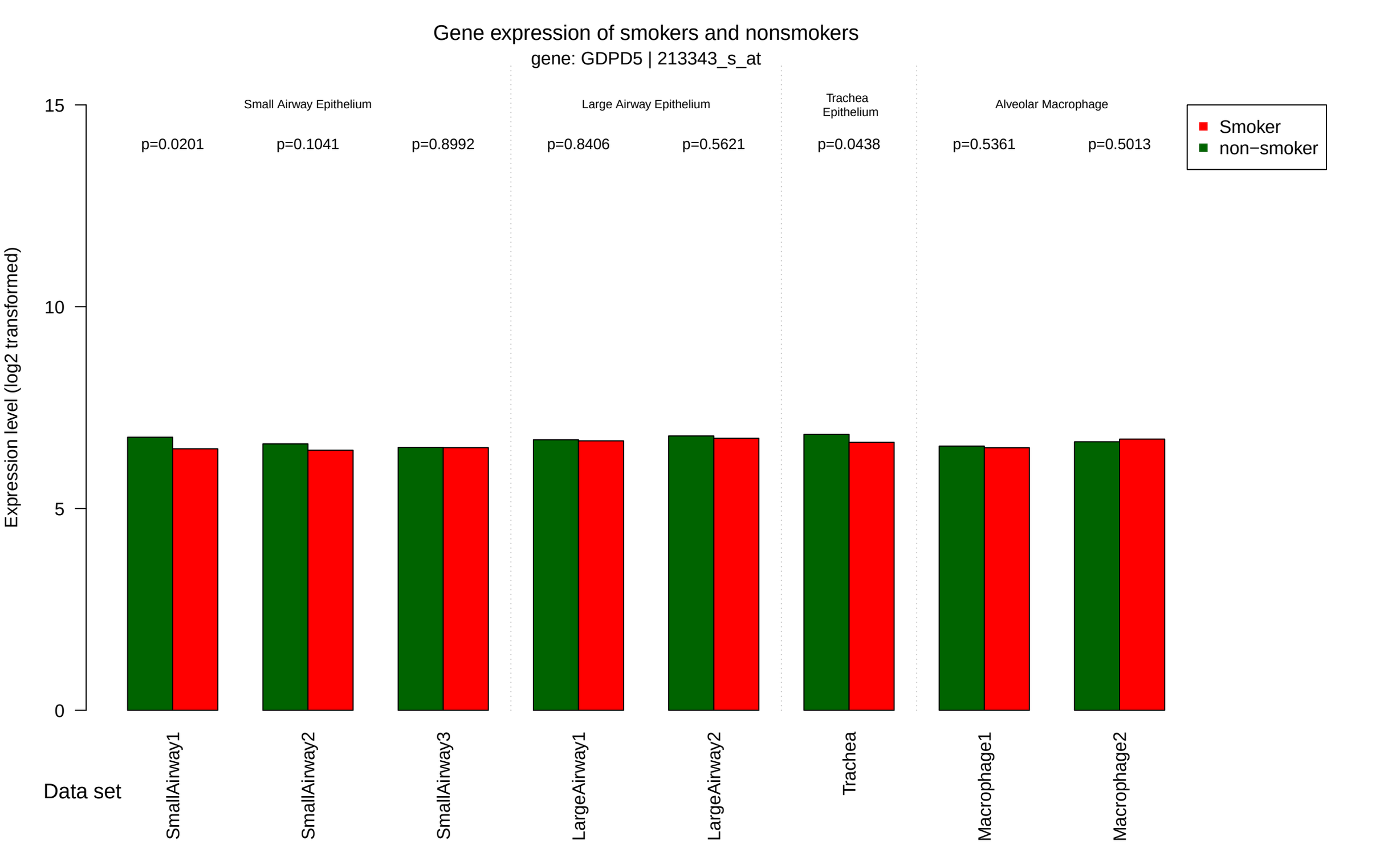
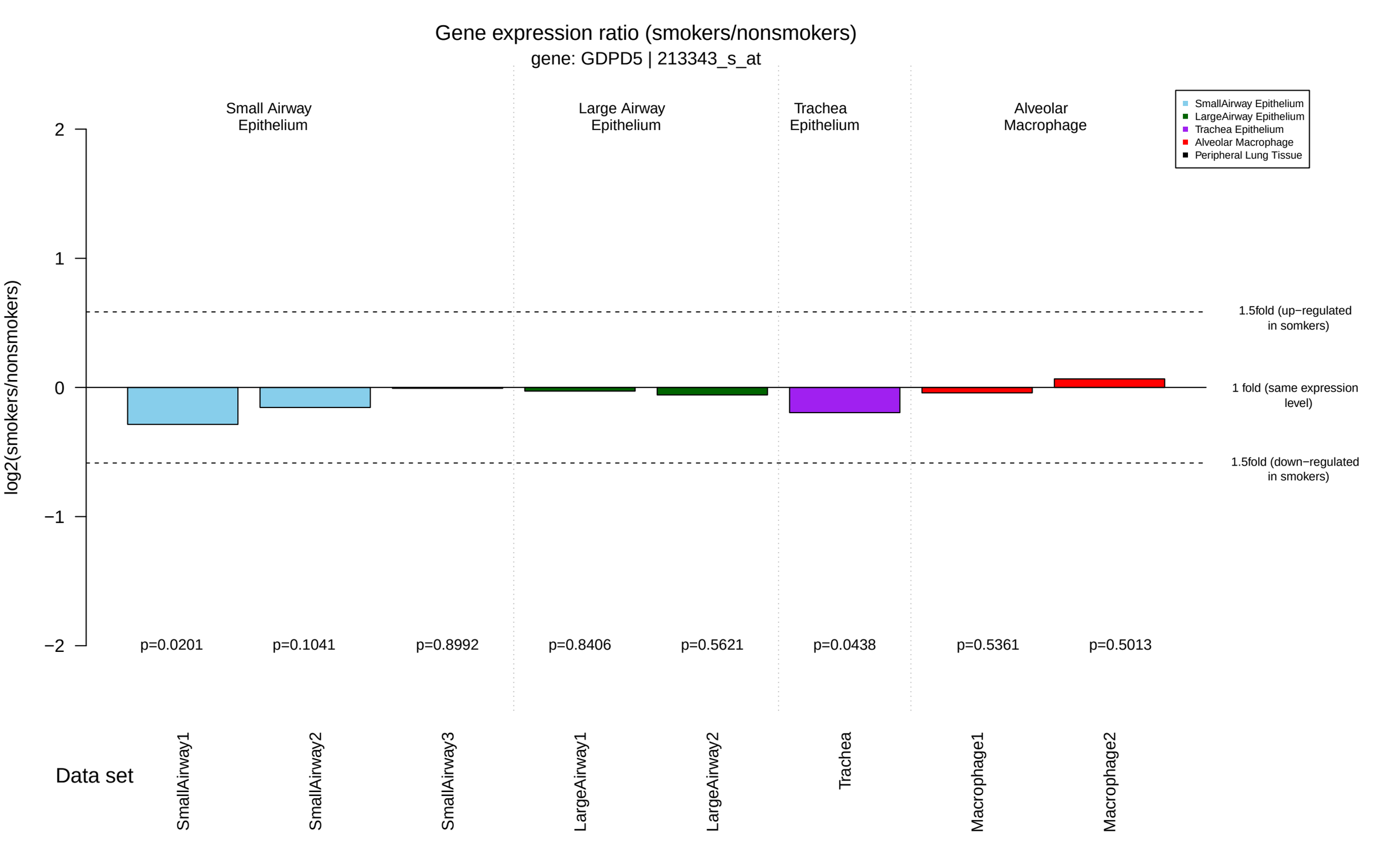
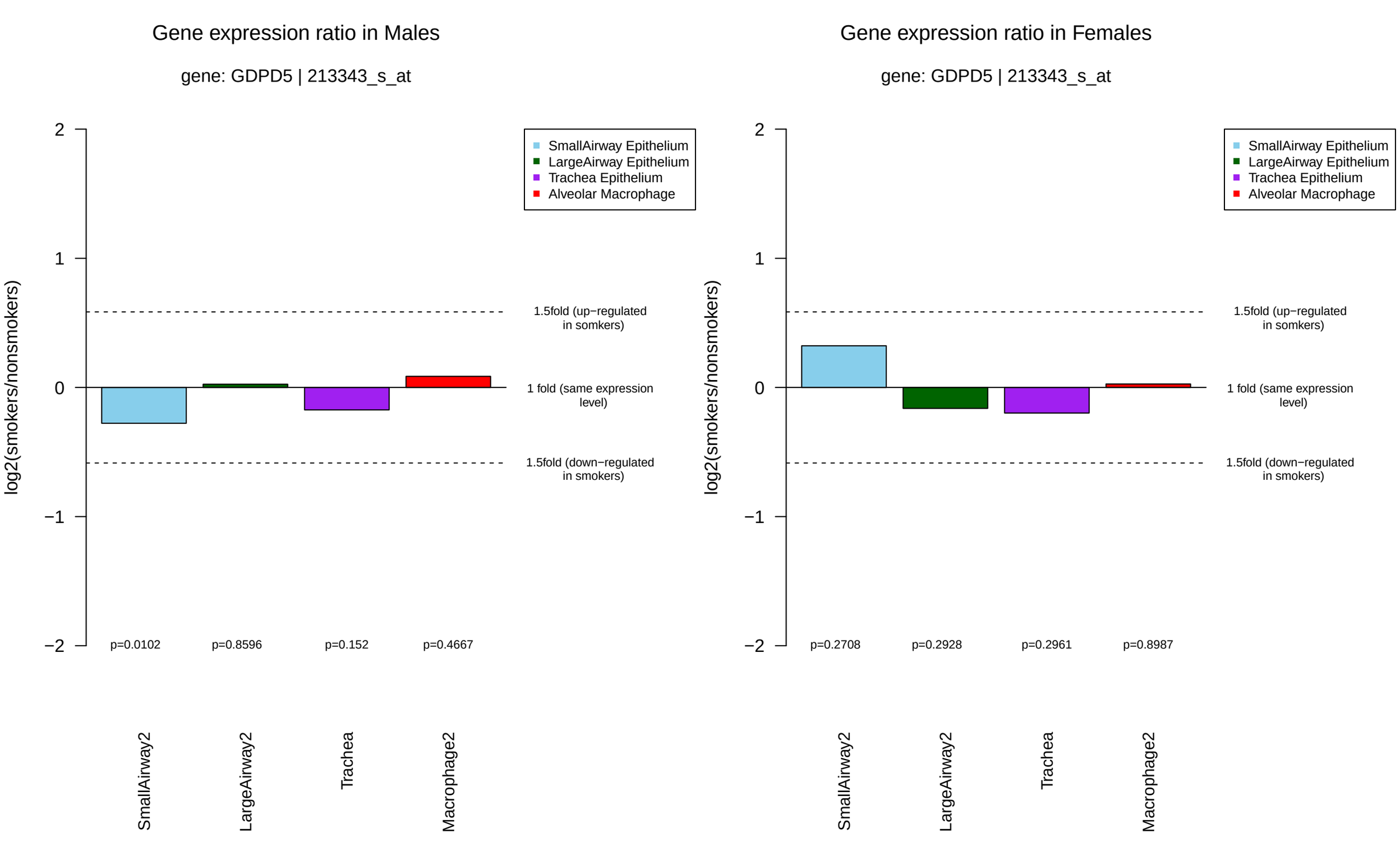
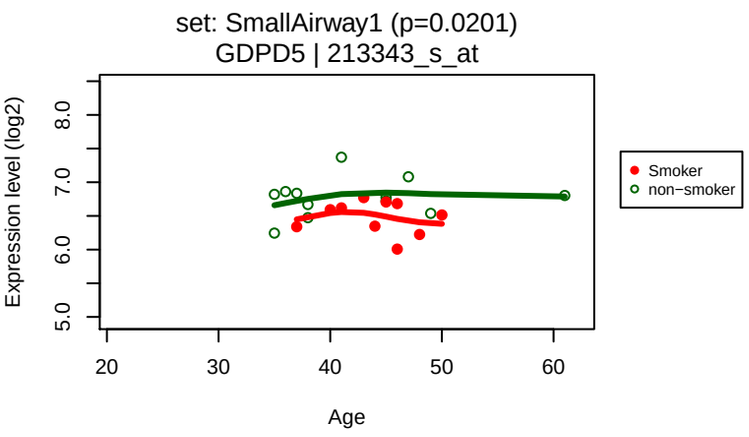
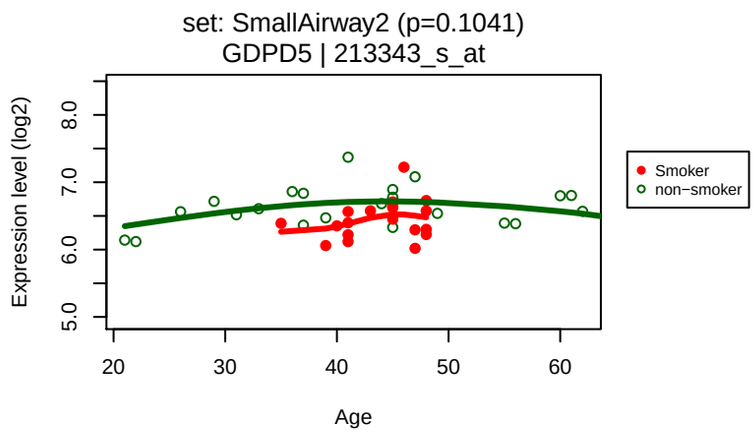
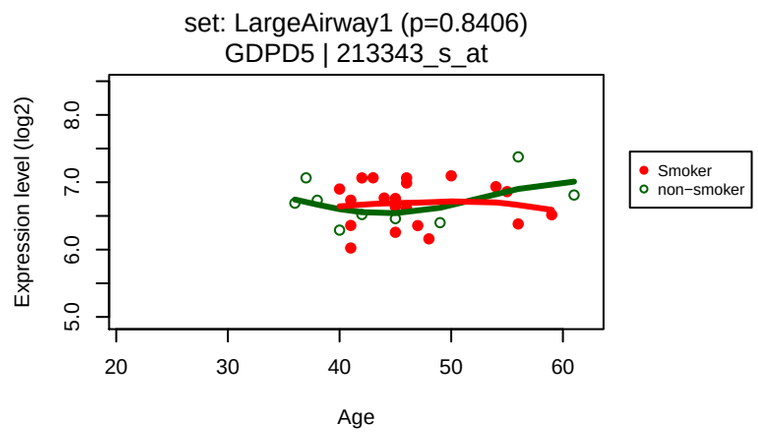

Gene expression of smokers and nonsmokers
gene: GDPD5 | 213343_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0201
p=0.1041
p=0.8992
p=0.8406
p=0.5621
p=0.0438
p=0.5361
p=0.5013
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

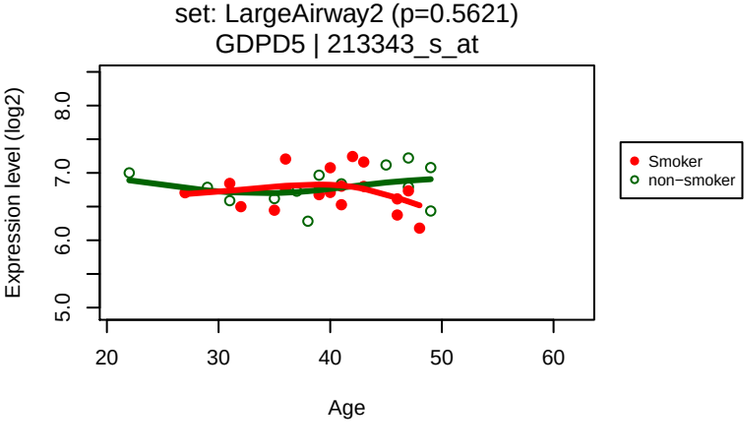

Gene expression ratio (smokers/nonsmokers)
gene: GDPD5 | 213343_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0201
p=0.1041
p=0.8992
p=0.8406
p=0.5621
p=0.0438
p=0.5361
p=0.5013
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

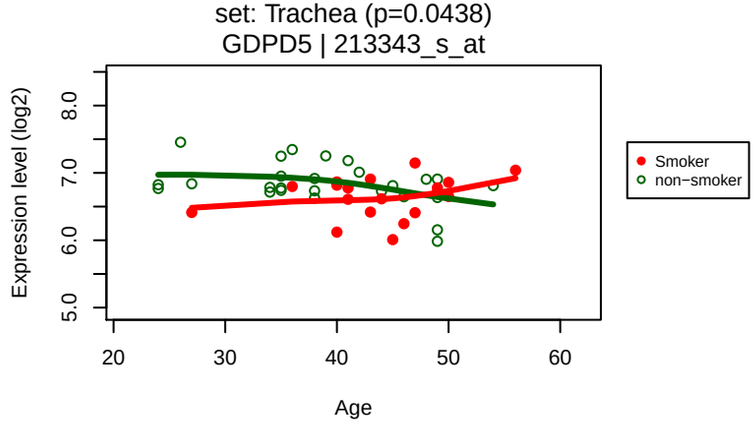

Gene expression ratio in Males
gene: GDPD5 | 213343_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0102
p=0.8596
p=0.152
p=0.4667
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GDPD5 | 213343_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2708
p=0.2928
p=0.2961
p=0.8987
SmallAirway2
LargeAirway2
Trachea
Macrophage2

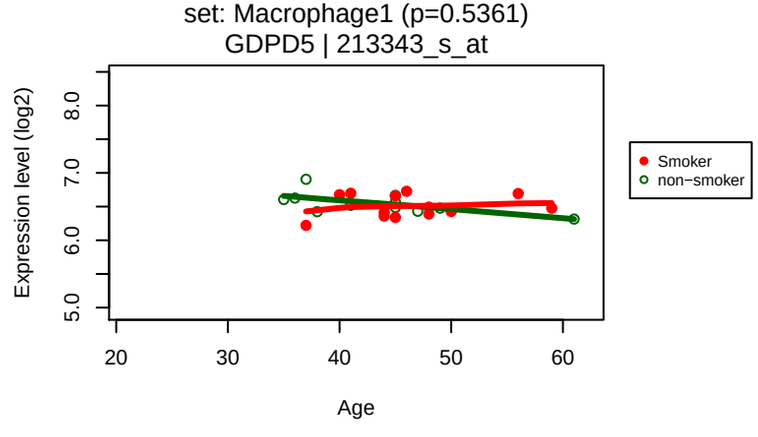

set: SmallAirway1 (p=0.0201)
GDPD5 | 213343_s_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker

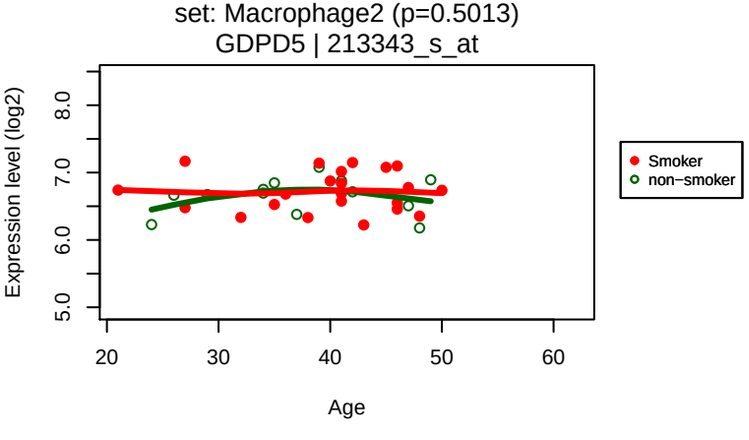

set: SmallAirway2 (p=0.1041)
GDPD5 | 213343_s_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker

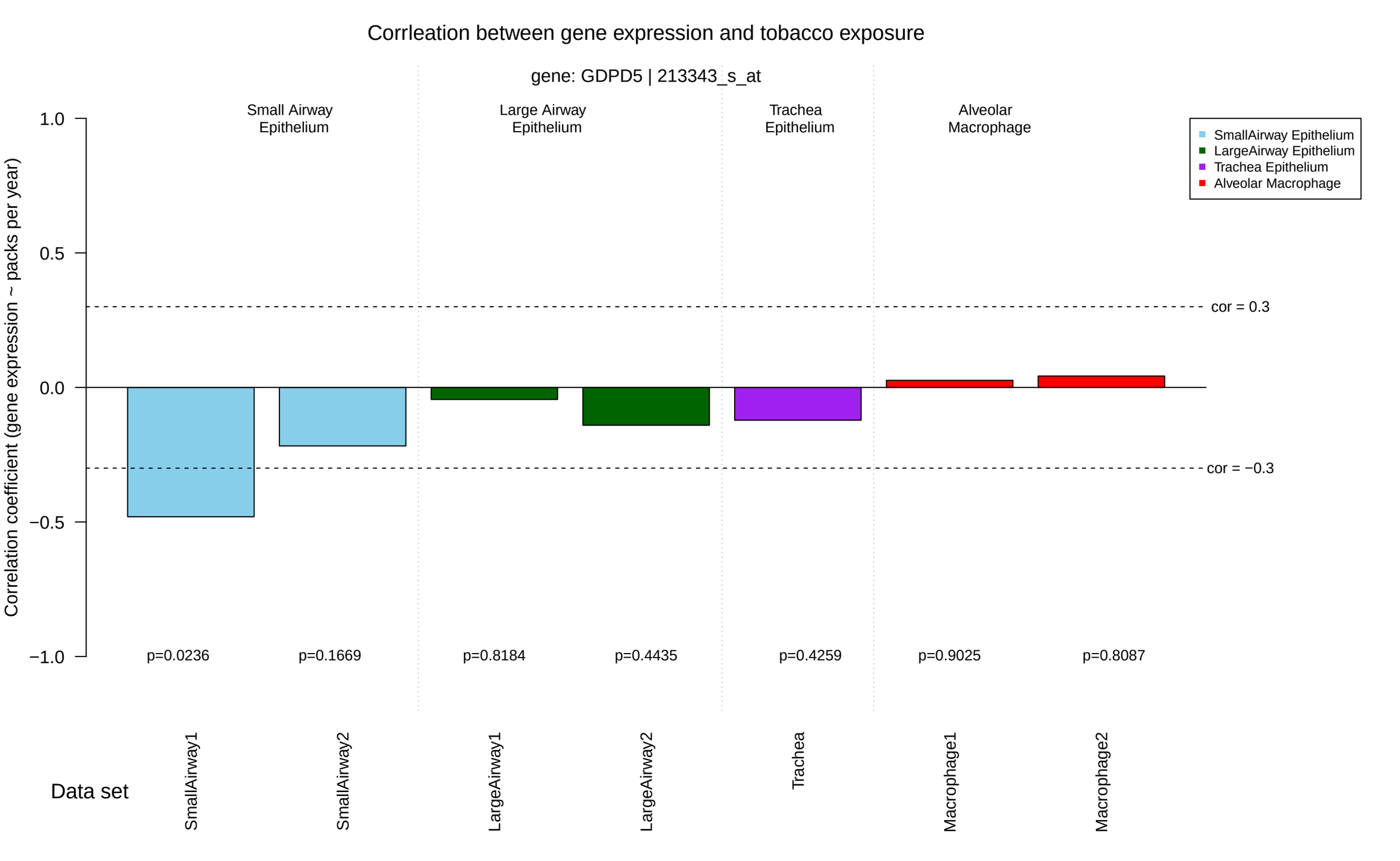

set: LargeAirway1 (p=0.8406)
GDPD5 | 213343_s_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5621)
GDPD5 | 213343_s_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0438)
GDPD5 | 213343_s_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5361)
GDPD5 | 213343_s_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5013)
GDPD5 | 213343_s_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GDPD5 | 213343_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0236
p=0.1669
p=0.8184
p=0.4435
p=0.4259
p=0.9025
p=0.8087
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2